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AVDASI-3: Sustainable Aviation & Ethics

Lecture-6: Sustainable Aerodynamics & Configurations

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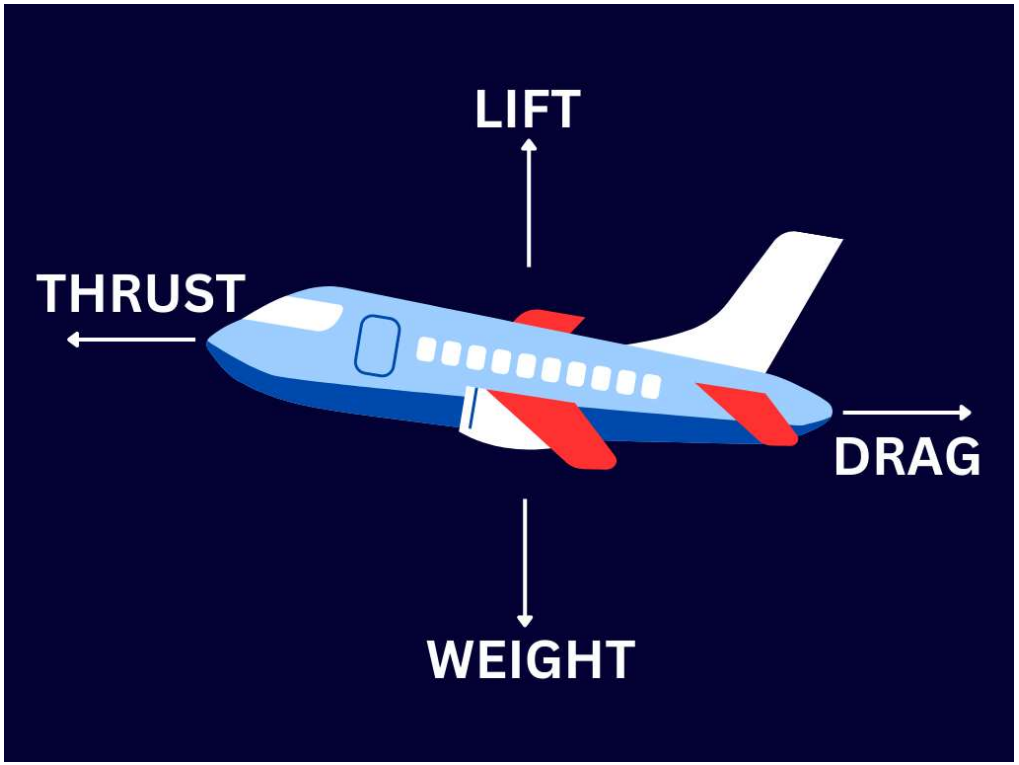
Simplified aircraft force balance

At level cruise

$$W = L = \tfrac{1}{2} \rho V^2 S C_L$$

$$T = D = \tfrac{1}{2} \rho V^2 S C_D$$



Aerodynamic efficiency index

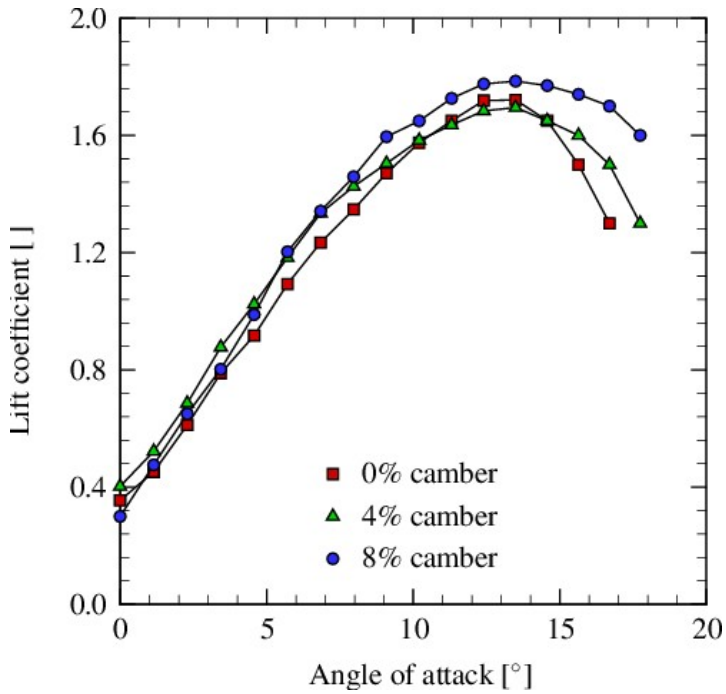
- Efficiency = $\dfrac{What\ I\ gain}{What\ I\ pay\ for\ that}$

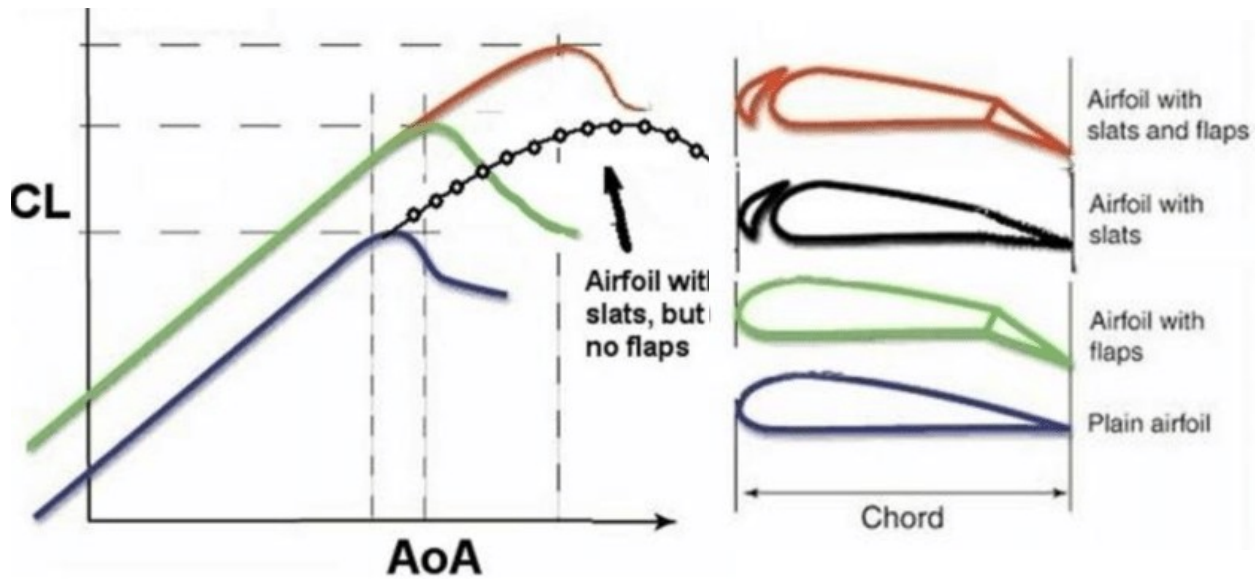
- Aerodynamic Efficiency = $\dfrac{Lift\ (L)}{Drag\ (D)} = \dfrac{C_L}{C_D}$

- Aerodynamic Efficiency$\uparrow \Rightarrow C_L\uparrow$ and / or $C_D\downarrow$

Increasing C_L

- High camber
- High lift devices





But…

In cruise (max time & energy spent by the a/c)

L = W = constant (approximation)

[in actual it's an opportunity to cruise climb as W reduces constantly]

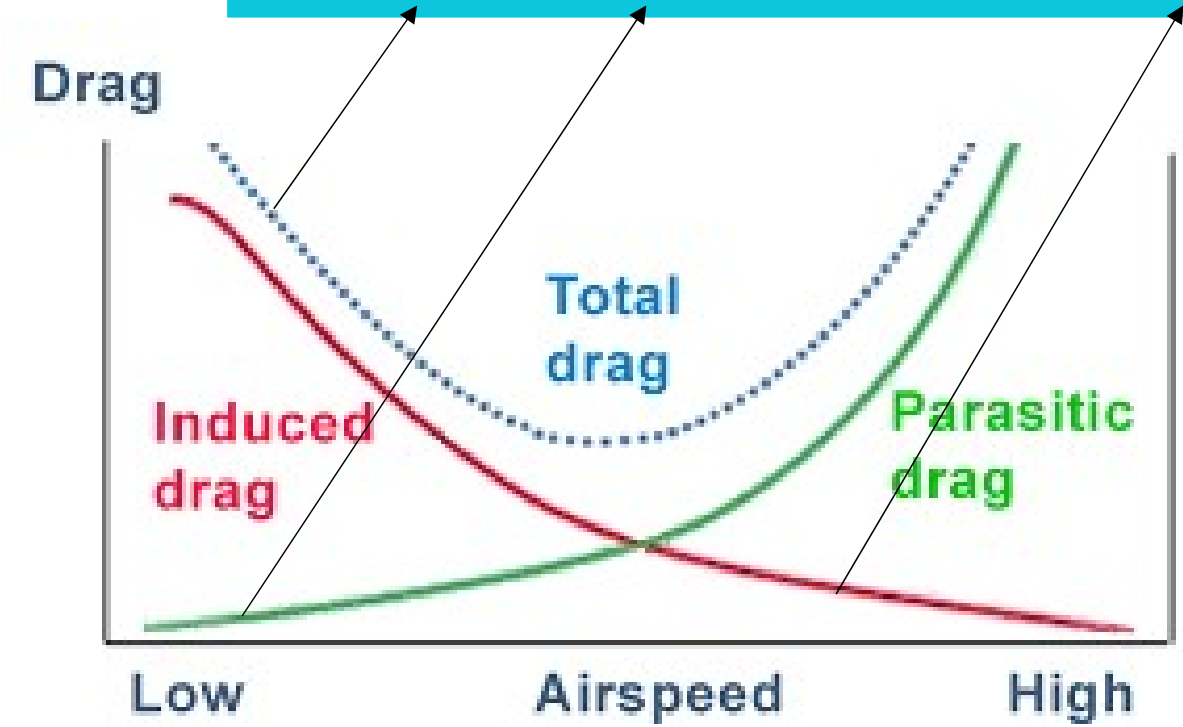
Hence

L / D = fn. of D (or C_D) only

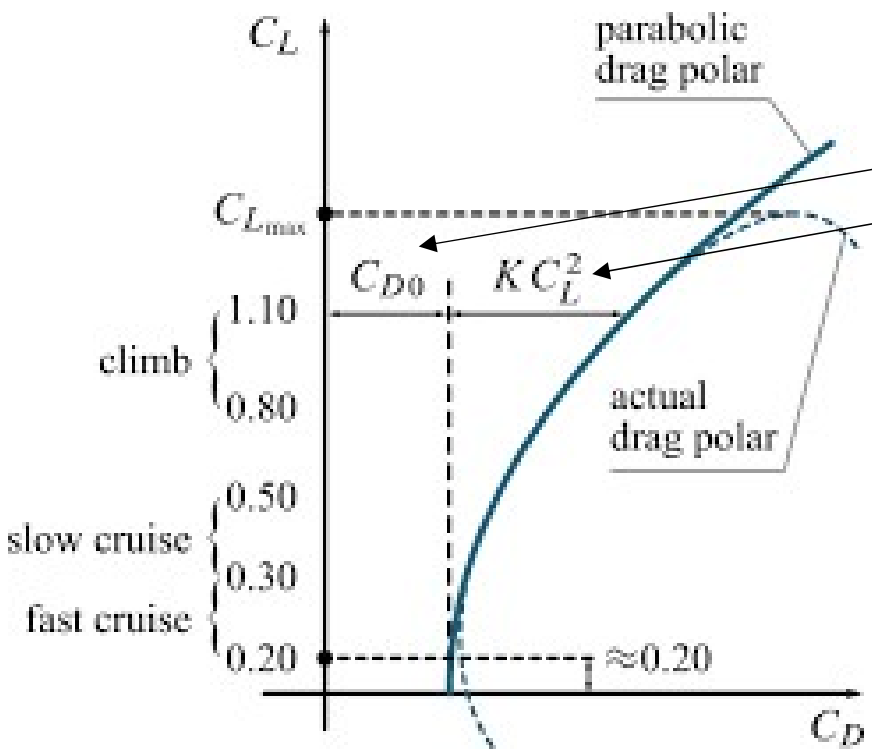
<u>Cruise Efficiency is a function of aircraft Drag only</u>

Total drag

Total Drag = D_0 (Form + Friction) + D_L (Lift induced)



Drag polar



Total Drag = D_0 + D_L

Parasite drag (zero lift or friction or form drag)

Induced drag (Lift induced)

$$C_D = C_{Do} + C_L^{\,2} / \pi A e$$

where:

C_D	is the drag coefficient, where the drag, $D = C_D S q$
C_{Do}	is the zero-lift drag coefficient (parasitic drag)
C_L	is the lift coefficient, where lift, $L = C_L S q$
A	is the wing aspect ratio, $A = b^2/S$
e	is Oswald's efficiency
S	is the wing reference area
q	is the dynamic pressure, $q = \rho v^2/2$
b	is the wing span
v	is the velocity

Reducing Drag

- Reducing D_0
  - Streamlined configuration
  - Smaller LGs, conformal antennae etc.
  - Smooth surface finish, flushed rivets / fasteners etc.
  - Low $\mathbf{S_{wet}}$
  - Technologies to delay flow separation

- Reducing D_L
  - Increase Aspect Ratio (i.e. slender wing)
  - Increase b_{eff} for same L or W

$$\mathbf{D_L = (L/b_{eff})^2 = (W/b_{eff})^2}$$

[b_{eff} = effective span = span + winglets]

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Novel aerodynamics to reduce drag

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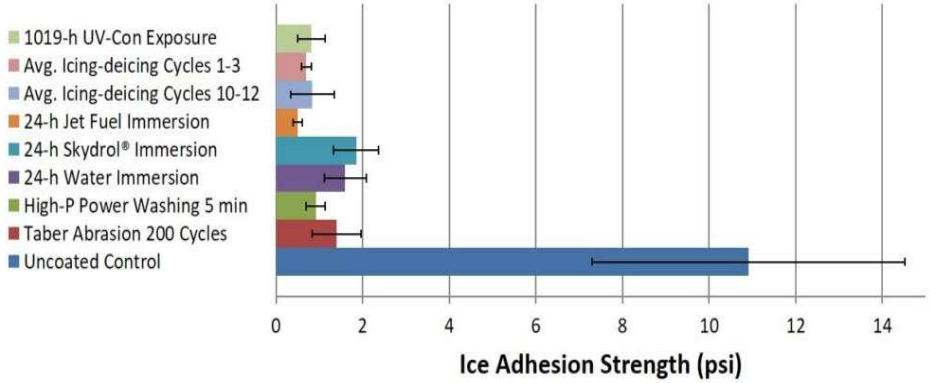
Boundary layer control

- Managing thin layer of air clinging to the a/c can significantly reduce drag (D_0↓)
  - Hybrid Laminar Flow Control (HLFC): Actively sucking in slow-moving boundary layer air to reduce drag
    - NASA tested HLFC on a Boeing 757 showing fuel savings of up to 15%
  - Coatings and Materials: Developing surfaces that reduce friction effectively
    - Anti-icing coatings
    - Bug-phobic coatings
      - Boeing/NASA's Environmentally Responsible Aviation (ERA) Project

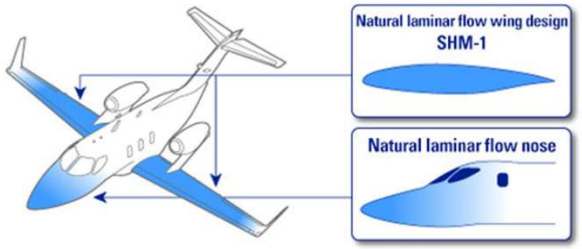
(PDF) Designing A Hybrid Laminar-Flow Control Experiment -- The Cfd-Experiment Connection

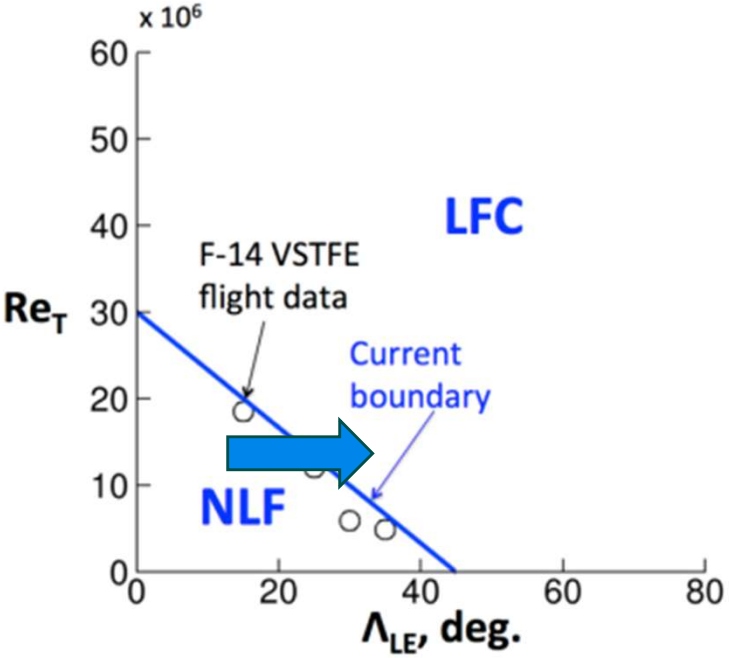


Anti-Ice Coating for Aerospace Industry | NEI Corporation

Natural laminar flow (NLF)

- Passive method for shape optimization to maintain smoother airflow & reduce drag (D_0↓)
- HondaJet
- NASA's X-59 QueSST
  - Quiet SuperSonic Technology
  - Allows for higher sweep at high speed (Re) w/o LFC





Design Innovation 2 | HondaJet | Products | Honda Global

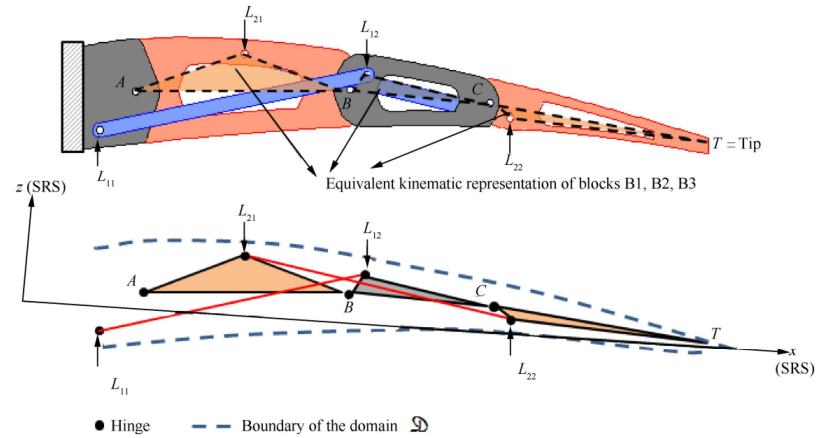
Bozeman2024_final.pdf

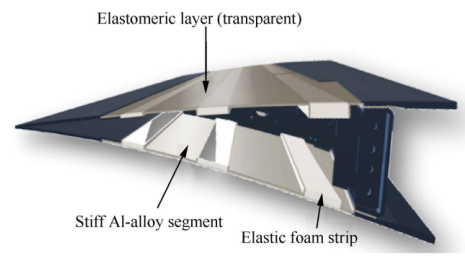
Riblets & surface textures

- Bio-mimicry: Inspired by sharkskin
- Tiny grooves on aircraft surface (molded into a plastic film with an adhesive backing) to reduce drag by controlling airflow ($D_0\downarrow$)
- Airbus tested riblet coatings on an A350, claiming a 1% fuel savings (significant when applied fleet-wide)

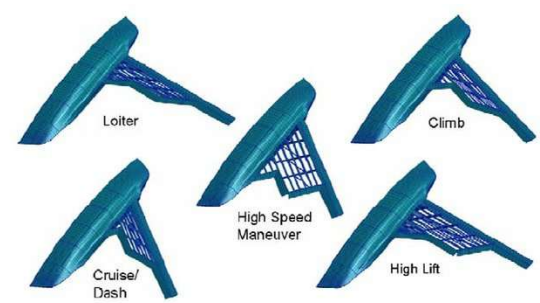
Biomimicry: engineering in nature's style | Airbus

Morphing / adaptive wing





[SARISTU: Adaptive Trailing Edge Device (ATED) design process review]



[(PDF) STRUCTURAL DESIGN AND ANALYSIS OF THE MISSION ADAPTIVE WINGS OF AN UNMANNED AERIAL VEHICLE]

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[At last, a shape-morphing aircraft wing is here - no flaps - YouTube]

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Novel aircraft configurations

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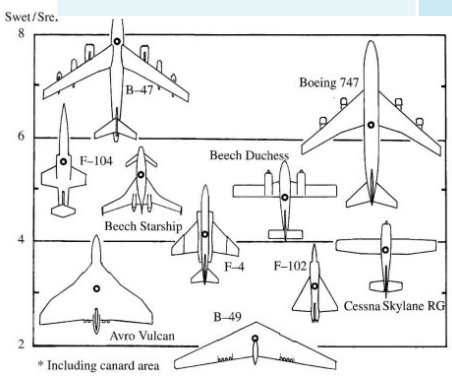
Truss based wing

- **Motivation:** Higher AR or b_{eff} (with optimal wt. penalty) $\Rightarrow D_L\downarrow$
- **Program details**
  - Name: X-66
  - Sponsor: NASA's Subsonic Ultra-Green Aircraft Reach (SUGAR) program
  - Developer: Boeing (POC with modified MD-90 for flight in 2028)
- **Platform details**
  - Slender wing design stabilized by diagonal bracing struts (Transonic Truss-Braced Wing) achieving Aspect Ratios up to 27
  - Optimized truss for 52 m span wing for flying up to Mach 0.80
  - Wing folds outboard of the truss for ICAO Code C (36 m) applications
  - Fuel burn 8-10% less than cantilevered wings; with better propulsion and materials, efficiency is targeted at 30% better than B-737 Max & A320neo

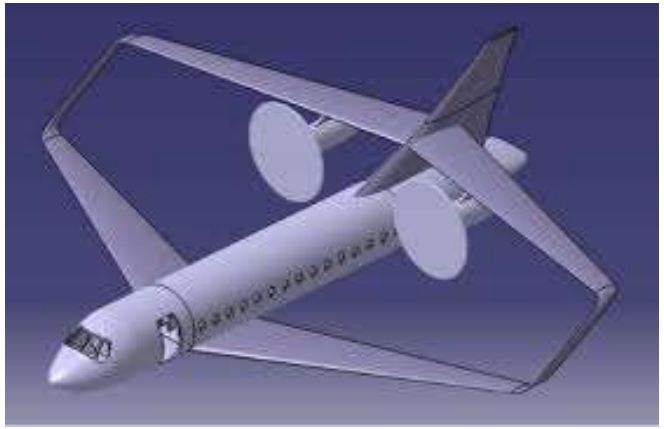
Blended wing (non-military)

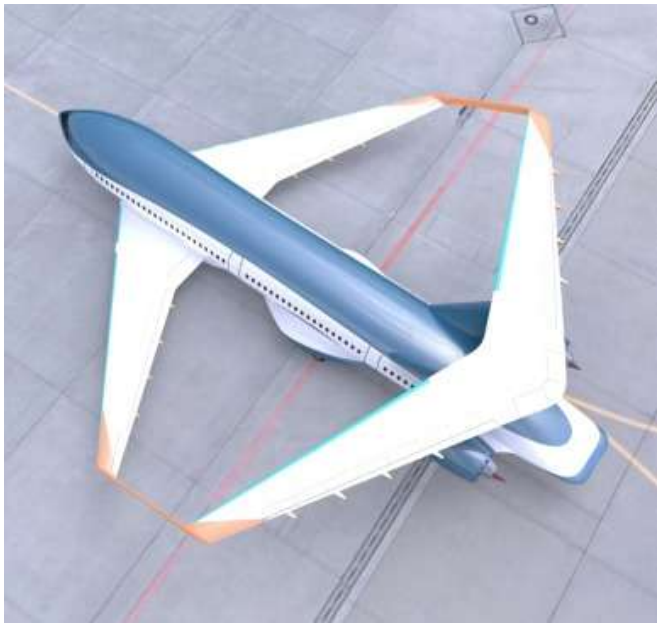
BWB Aircraft	Airbus MAVERIC	BOEING X-48	JET-ZERO BWB
Motivations	Low S_{wet} for given volume & minimum surface discontinuities $\Rightarrow D_0\downarrow$ Winglets to increase $b_{eff} \Rightarrow D_L\downarrow$ Distributed weight leading to lower structural weight $\Rightarrow$ W & L & $D_L\downarrow$		
Platform details	• Fuel burn @ 20% less • Scaled POC, flown in 2019	• Scaled POC, flown in 2012 • Built @ Cranfield Aero., UK	• Fuel burn target @50% • Target launch in 2030 • 100% SAF
	Major concerns: Controls, passenger evacuation, engine failure case		



Prandtl wing

- **Motivation:** Higher b_{eff} (with optimal wt. penalty) $\Rightarrow D_L\downarrow$
- **Program details**
  - Name: PARSIFAL (Prandtlplane ARchitecture for the Sustainable Improvement of Future AirpLanes) – Academic project
  - Sponsor / Developer: University of Pisa
- **Platform details**
  - Closed wing system of 2 main wings connected to each other by side wings at the tips (like huge winglets), sweep for transonics
  - Least induced drag (no tip vortex) for given span & weight leading to 10-20% less fuel burn
  - Redundant C/S for direct lift control (change lift @ same AoA)
  - Same airport infrastructure, but improved efficiency for lower D_L





Words of caution…

- Snowball effect from any unconventional design
- Manufacturability
- Certification
- Passenger safety / comfort
- Airport integration

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Thank you

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